Transfer learning in audio-based emotion detection

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Original Project Reference: Language Model Fine-Tuning with Triplet Loss for Improved Sentence Embeddings

Abstract: We propose an improvement to the Stanford CS224n project that fine-tuned BERT with triplet loss to generate higher-quality sentence embeddings.

Our innovation introduces dynamic hard negative mining during training to automatically select more challenging negative samples instead of random negatives. We hypothesize that this approach will result in embeddings that better distinguish fine-grained semantic differences, leading to superior performance on sentence similarity, retrieval, and clustering tasks.

Introduction: Learning high-quality sentence embeddings is crucial for many NLP applications like semantic search, question answering, and clustering. Although the original Stanford project improved embeddings with triplet loss, it relied on random or static negative samples. We propose dynamic hard triplet mining, where during training, the hardest negatives (most similar but incorrect) are selected dynamically, forcing the model to learn more robust embeddings.

Proposed Approach

Baseline: Fine-tune pre-trained BERT using traditional triplet loss (anchor, positive, random negative).

Our Method:
- Use **dynamic mining** during training to select the hardest negative (closest to the anchor among non-matching sentences) in each batch.
- Implement an efficient mining mechanism using cosine similarity between embeddings.
- Fine-tune using triplet loss with these dynamically selected negatives.

Expected Impact:

Stronger contrast between positives and negatives.

Faster and more stable convergence.

Better generalization on unseen data.

Expected Results

We expect:

- Higher sentence similarity correlation.
- Better retrieval results.
- More meaningful clustering in embedding space.

Potential Challenges

- Selecting hard negatives dynamically might increase computation.
- Risk of **over-mining** (only focusing on extremely difficult samples).

Innovation Summary

- The original project used random negatives.
- We will introduce dynamic hard negative mining, selecting smarter negatives on-the-fly.
- This innovation is inspired by successful hard mining techniques in computer vision, applied here in NLP